Weather-Driven Demand Forecasting and Workforce Optimization: Leveraging Predictive Analytics for Strategic Resource Management

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Abstract

This capstone project examines whether weather-sensitive demand forecasting can improve forecast accuracy and workforce allocation efficiency in Ontario's retail sector. Traditional forecasting methods often rely mainly on historical demand patterns and may overlook external factors such as temperature, precipitation, snowfall, and solar conditions that can influence consumer behavior.

The study was conducted in two phases. In Phase 1, monthly Ontario retail sales data was combined with weather data from the NASA POWER database. Weather features were engineered and incorporated into four forecasting models: Naïve, Seasonal Naïve, SARIMA, and SARIMAX. Model performance was evaluated using a rolling-origin expanding-window approach with Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Diebold-Mariano test. In Phase 2, forecast outputs were converted into staffing requirements using public employment and wage data, allowing workforce mismatch and staffing penalty cost to be compared under baseline and weather-sensitive scenarios.

The results showed that weather variables did not improve forecasting uniformly across sectors. Among the five selected sectors, SARIMAX produced a meaningful and statistically significant improvement in only one case: Other motor vehicle dealers [4412]. That same sector also showed lower workforce mismatch and staffing penalty cost. Overall, the study provides partial support for both research questions and demonstrates a practical framework linking weather data, demand forecasting, and workforce planning.

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1. Introduction

1.1. Background and Context

Accurate demand forecasting is a fundamental component of effective operational planning. Organizations across retail and service industries rely on demand forecasts to guide critical decisions related to inventory management, staffing levels, and resource allocation. Traditionally, many forecasting systems rely primarily on historical demand patterns to estimate future activity. While historical trends can provide useful signals, they often fail to capture the influence of external environmental factors that can significantly alter customer behavior and demand patterns (Auramo, 2020).

Among these external factors, weather conditions play a particularly important role. Temperature fluctuations, precipitation levels, snowfall, cloud amount, and other extreme weather events can influence consumer mobility, purchasing behavior, and overall economic activity. For example, as heavy snowfall or rainfall may reduce in-store visits, while unusually warm temperatures can increase demand for certain seasonal products (Arunraj & Ahrens, 2016). Despite these observable patterns, many organizations continue to rely on models that consider only historical sales data, which can limit forecasting accuracy when demand is influenced by environmental variability (Auramo, 2020).

A review of publicly available meteorological records on sources such as Statistics Canada and NASA POWER reveals that the volume of weather data has grown considerably in recent years. This growth, combined with advances in statistical modeling, creates new opportunities to incorporate external variables into forecasting frameworks. Weather-sensitive forecasting models attempt to integrate environmental indicators alongside historical demand signals, allowing

organizations to better anticipate demand fluctuations that arise from changing weather conditions.

Beyond forecasting accuracy, improved demand predictions can also influence downstream operational decisions, particularly workforce planning (Adebisi et al., 2025). Workforce allocation decisions are often based on expected demand levels; therefore, inaccurate forecasts can lead to operational inefficiencies. Underestimation of demand can result in understaffing, longer wait times, and reduced service quality, while overestimation may lead to unnecessary labor costs and idle staff. Consequently, improving forecast accuracy has the potential to directly enhance workforce allocation efficiency.

This capstone project investigates the role of weather conditions as an external driver of demand and evaluates whether incorporating weather variables into statistical forecasting models can improve both forecast accuracy and workforce allocation decisions. By combining historical retail demand data with meteorological information, the study explores whether weather-sensitive demand forecasts provide meaningful improvements over traditional forecasting approaches and whether these improvements translate into more efficient staffing strategies.

1.2. Problem Identification

Many organizations base demand forecasting and workforce planning decisions primarily on historical demand trends. While this approach can capture long-term patterns and seasonal effects, it often overlooks the influence of external environmental variables that may cause sudden fluctuations in demand (Auramo, 2020). Weather conditions such as temperature changes, rainfall, or snowfall can significantly affect consumer behavior, particularly in retail sectors where purchasing activity is sensitive to outdoor conditions (Arunraj & Ahrens, 2016).

When forecasting models ignore these external drivers, we believe organizations may experience inaccurate demand predictions that lead to operational inefficiencies. Forecasting errors can translate into workforce allocation mismatches, where staffing levels do not align with actual demand. Overstaffing increases operational costs by allocating more labor resources than necessary, while understaffing may reduce service quality, increase customer wait times, and negatively affect overall operational performance (Adebisi et al., 2025).

As climate variability increases and weather patterns become more unpredictable, the limitations of traditional forecasting models become more pronounced. Incorporating weather variables into forecasting frameworks may provide additional predictive power by capturing external influences that historical demand alone cannot explain (Babongo et al., 2018). However, the extent to which weather information improves forecasting accuracy, and whether such improvements lead to more efficient workforce allocation, remains an important empirical question.

This project addresses this gap by developing a weather-sensitive demand forecasting framework and evaluating its potential impact on workforce planning efficiency.

1.3. Research Questions

This study is guided by the following research questions:

1. How does the inclusion of weather indicators affect the accuracy of statistical demand forecasts in different retail subsectors?
2. How effectively can weather-sensitive demand forecasts improve workforce allocation efficiency in these retail subsectors?

1.4. Hypotheses

Two hypotheses are tested in this project to evaluate the impact of weather-sensitive forecasting on demand prediction and workforce planning.

1.4.1. Forecast Accuracy Hypothesis

H_0: Incorporating weather-related variables into statistical demand forecasting models does not result in a statistically significant improvement in forecast accuracy compared to traditional forecasting methods.

H_1: Incorporating weather-related variables into statistical demand forecasting models results in a statistically significant improvement in forecast accuracy compared to traditional forecasting methods.

1.4.2. Workforce Allocation Efficiency Hypothesis

H_0: Workforce allocation decisions based on weather-sensitive demand forecasts do not significantly reduce workforce mismatches (overstaffing or understaffing) compared to traditional forecasting methods.

H_1: Workforce allocation decisions based on weather-sensitive demand forecasts significantly reduce workforce mismatches (overstaffing or understaffing) compared to traditional forecasting methods.

2. Data Collection and Preparation

2.1. Data Sources

This project integrates three primary categories of data: retail demand data, meteorological data, and workforce data. The objective of combining these datasets is to evaluate whether weather-related factors contribute to improvements in demand forecasting accuracy and workforce allocation decisions in several retail subsector between 2017 and 2025.

Retail demand data was obtained from publicly available tables provided by Statistics Canada (Appendix A) reporting monthly retail trade activity in Ontario, Canada. The dataset contains aggregated monthly sales values across multiple retail sectors classified according to the North American Industry Classification System (NAICS). In total, the dataset includes twenty-eight distinct retail sectors, covering a wide range of consumer-facing industries such as motor vehicle dealers, furniture stores, electronics retailers, clothing stores, and food services.

Meteorological data was obtained from the NASA POWER database through its public application programming interface (API). NASA POWER provides historical daily weather observations and derived meteorological variables for locations worldwide. The dataset includes several weather indicators relevant to consumer behavior and operational demand, including mean temperature, maximum temperature, minimum temperature, precipitation levels, snowfall estimates, cloud coverage, wind speed, and solar radiation. These variables capture environmental conditions that may influence customer mobility, shopping behavior, and seasonal purchasing patterns.

Finally, workforce data tables were collected from Statistics Canada, these datasets provided trade-level information related to labor cost structures and employment characteristics necessary to translate forecasted retail demand into staffing requirements.

2.2. Data Engineering

The weather dataset was constructed using a custom data extraction pipeline implemented in Python. The NASA POWER API was used to programmatically download historical daily weather data for multiple geographic coordinates across Ontario. The extraction process covered the period between 2017 and 2025, consistent with the available retail demand data and workforce data, ensuring that all datasets could be merged for joint analysis.

To represent weather conditions across the province, a geographic bounding box covering Ontario was defined using latitude and longitude ranges. Within this bounding box, 35 sampling points were generated to capture spatial variation in weather conditions. For each sampling location, daily weather variables were requested from the NASA POWER API and stored as structured records.

After downloading the weather observations for all grid points, the data was combined into a single dataset. Daily weather values were averaged across all sampled locations to create a province-wide daily weather representation. This aggregation approach reduces localized variability while preserving the overall environmental conditions experienced by consumers across the entire province.

Once daily weather values were obtained, the data was transformed into monthly aggregates to align with the temporal frequency of the retail demand data. Monthly features were generated by computing averages or totals depending on the nature of the variable. Temperature-related variables were averaged across days within each month, while precipitation-related variables were aggregated using monthly sums where appropriate. The initial goal was to conduct this analysis at the daily level; however, due to the limited availability of public demand data, aggregating to a monthly level was necessary to proceed.

The resulting monthly weather dataset contained multiple environmental indicators representing conditions that could potentially influence retail demand, suitable for use as explanatory variables in statistical forecasting models.

2.3. Data Cleaning and Preparation

Prior to conducting statistical analysis, the retail demand dataset, the weather dataset, and the workforce datasets underwent several preprocessing steps to ensure data consistency, reliability, and compatibility.

First, the datasets were examined for missing values and inconsistencies. Any missing or incomplete observations were identified and evaluated to determine whether imputation or exclusion was necessary. Because the datasets were derived from structured public sources, the frequency of missing values was limited. Where minor gaps occurred in the weather data, standard aggregation methods ensured that monthly summaries remained consistent.

Next, the datasets were aligned temporally. Retail demand data was available at a monthly frequency, while the weather dataset was initially collected at a daily level. To ensure comparability, the daily weather data was aggregated to monthly observations using statistical summaries appropriate to each variable as discussed in the previous section. This transformation allowed both datasets to share the same time index.

Additional preprocessing steps included formatting date variables, ensuring consistent numeric types, and organizing the datasets into structured data frames suitable for statistical modeling in Python. Retail sector identifiers were also standardized to ensure consistency across the analysis pipeline.

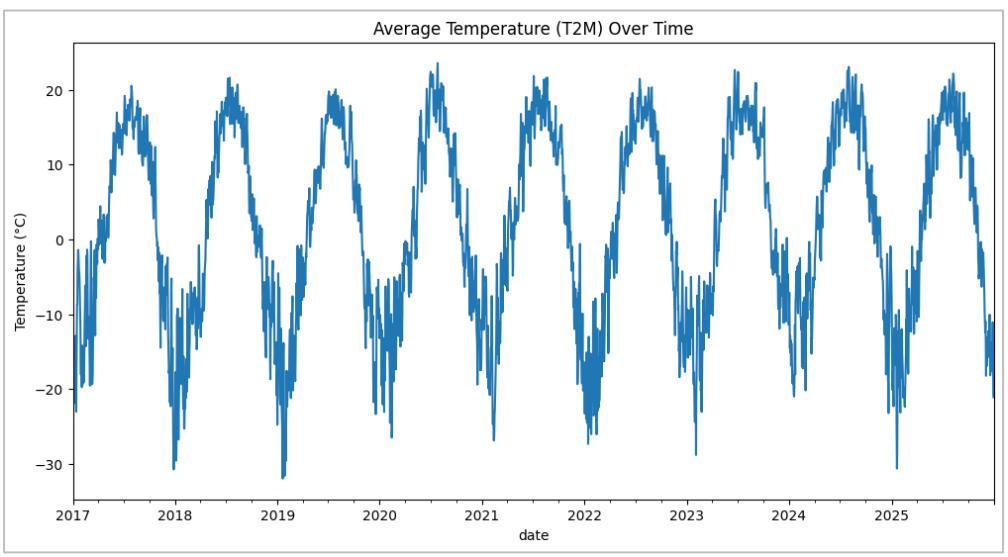
Finally, the weather dataset and retail demand dataset were merged using their common monthly time index. The resulting integrated dataset contains both demand indicators and environmental variables for each time period, enabling the subsequent analysis of weather-sensitive forecasting models.

2.4. Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to better understand the characteristics of both the weather dataset and the retail demand data prior to model development. The objective of this stage was to identify patterns, detect anomalies, and evaluate potential relationships between weather conditions and retail demand.

Initial analysis focused on examining the distribution and seasonal patterns of key weather variables such as temperature and precipitation. Visualizations were used to observe long-term trends, seasonal cycles, and variability across the observation period. As expected, temperature variables showed strong seasonal behavior corresponding to typical winter and summer climate patterns in Ontario (Figure 1).

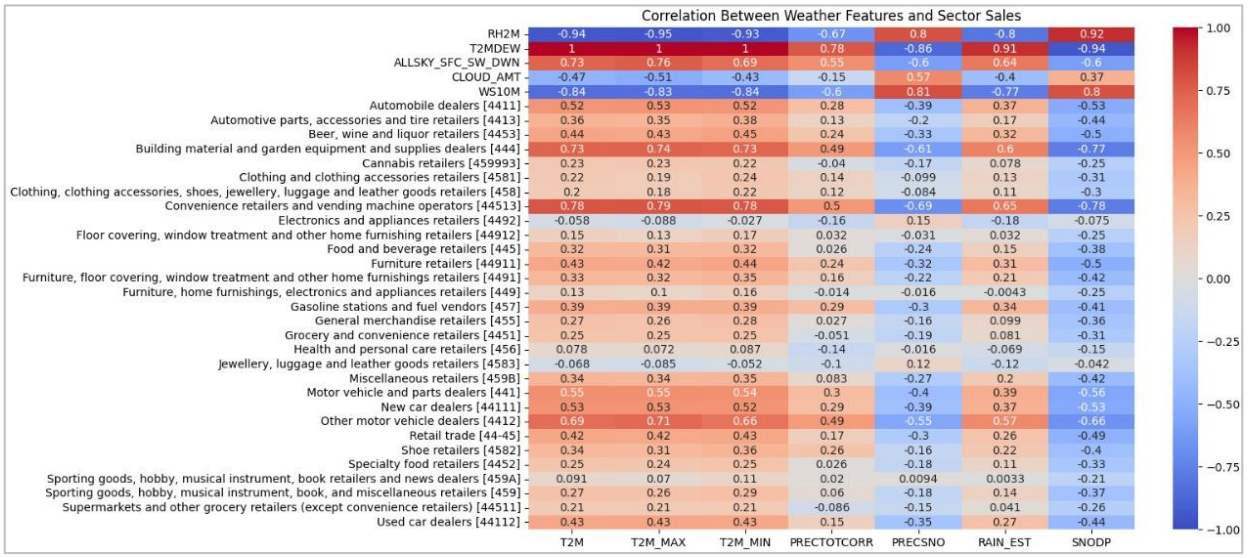
Figure 1. *Average Temperature (T2M) Over Time (2017–2025).*



Correlation analysis was performed to explore potential relationships between weather indicators and retail sales across sectors (Figure 2). While many sectors exhibited limited sensitivity to weather variables, preliminary analysis suggested that certain sectors may

demonstrate stronger relationships with environmental conditions. These findings informed the subsequent selection of sectors for more detailed forecasting analysis.

Figure 2. *Correlation between retail subsectors sales and weather features.*



The insights obtained from exploratory analysis helped guide feature selection and modeling decisions in the forecasting phase of the project. By identifying potential weather-demand relationships and seasonal structures in the data, this stage provided a foundation for developing weather-sensitive statistical forecasting models in the next phase of the study.

3. Demand Forecasting Analysis (Phase 1)

3.1. Forecasting Framework

The first analytical phase of this study focuses on evaluating whether the inclusion of weather-related variables improves the accuracy of statistical demand forecasts. While typical methods of demand forecasting using only historical data can capture recurring demand cycles, it may fail to account for external environmental factors that influence consumer behavior such as weather conditions. The forecasting framework developed in this study aims to determine

whether incorporating weather indicators into statistical models provides measurable improvements in forecast performance.

To evaluate the contribution of weather variables, two types of time series models were implemented and compared. The first model serves as a baseline forecasting approach and relies exclusively on historical demand information. This model is specified using a Seasonal Autoregressive Integrated Moving Average (SARIMA) structure, which is commonly used for modeling time series data that exhibit both trend and seasonal patterns. SARIMA models capture temporal dependencies by incorporating autoregressive terms, moving average components, and seasonal differencing. By relying solely on historical demand observations, the SARIMA model provides a benchmark against which the performance of weather-informed models can be evaluated.

The second model extends the baseline framework by incorporating exogenous weather variables into the forecasting process. This specification is implemented using a Seasonal Autoregressive Integrated Moving Average model with exogenous regressors (SARIMAX). In this formulation, weather indicators are included as external explanatory variables alongside the autoregressive components of the time series model. The SARIMAX framework allows the forecasting model to capture both internal temporal dependencies and external environmental influences that may contribute to variations in retail demand.

The comparison between SARIMA and SARIMAX models forms the basis of the evaluation strategy for this phase of the project. If weather variables provide additional predictive power, the SARIMAX models should demonstrate improved forecast accuracy relative to the baseline SARIMA models. Conversely, if weather variables do not contribute meaningful information, both model types should exhibit similar predictive performance. By systematically comparing

these two modeling approaches across selected retail sectors, the analysis aims to determine whether weather-sensitive forecasting models offer measurable advantages over traditional demand forecasting techniques.

3.2. Feature Engineering and Weather Variable Selection

To incorporate environmental conditions into the forecasting models, several weather-related variables were engineered from the aggregated monthly weather dataset. The objective of this step was to construct weather indicators that could plausibly influence retail demand while avoiding excessive multicollinearity among predictors. Because weather variables often exhibit strong correlations with each other, careful feature construction and selection were necessary to ensure that the forecasting models remained stable and interpretable.

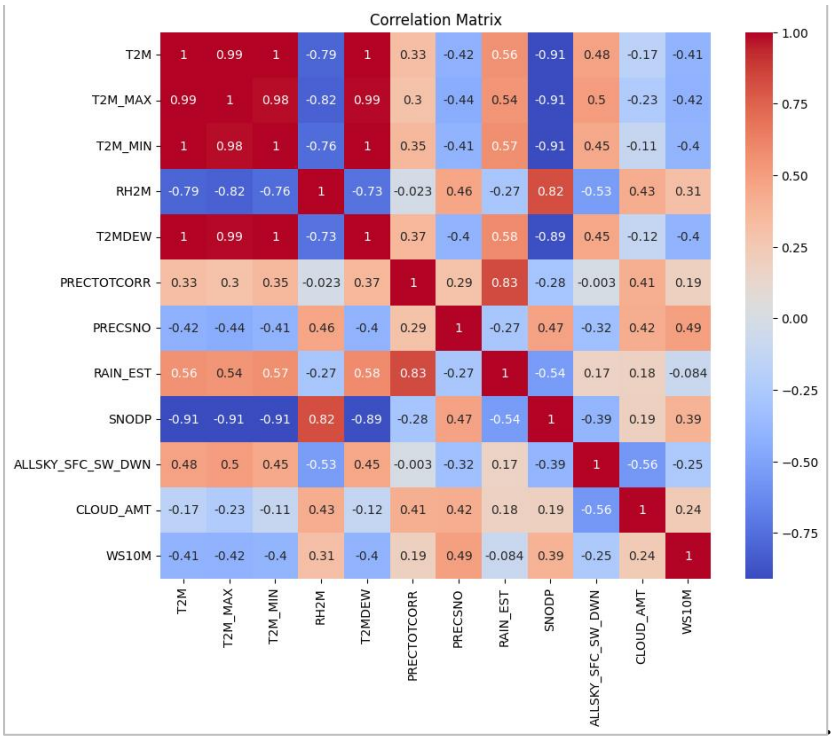
One of the primary engineered features used in the analysis is the concept of degree days, which are widely applied in economic and energy demand modeling to represent temperature-related deviations from a reference comfort (U.S. Environmental Protection Agency, 2021). Following the standard applied in Canadian climate measurement, an 18°C baseline was used to calculate two degree-day indicators: **Heating Degree Days (HDD)** and **Cooling Degree Days (CDD)** (Environment and Climate Change Canada, n.d.). Heating Degree Days measure the extent to which average temperatures fall below this baseline, indicating conditions that may increase heating-related activities or influence seasonal consumer behavior. Conversely, Cooling Degree Days capture the degree to which temperatures exceed the same baseline, representing warmer conditions that may affect mobility patterns, seasonal product demand, or outdoor activity levels. These indicators provide a more behaviorally meaningful representation of temperature variation than raw temperature values alone.

In addition to temperature-based indicators, other weather variables were incorporated to capture environmental conditions that could influence retail activity. These include **total precipitation**, which reflects rainfall intensity and may reduce consumer travel or in-store visits, and **solar radiation**, which serves as a proxy for general weather conditions such as brightness and cloud coverage that may affect outdoor activity levels. A binary **snow day indicator** was also created to identify months in which snowfall events occurred, representing weather conditions that could disrupt transportation or alter consumer behavior patterns.

Because several weather indicators may convey overlapping information, a multicollinearity diagnostic was performed to ensure that the set of explanatory variables used in the forecasting models did not introduce instability (Figure 3). Variance Inflation Factor (VIF) analysis was applied to evaluate the degree of correlation among candidate weather predictors. Variables exhibiting excessively high VIF values were removed to prevent redundancy and reduce the risk of inflated parameter estimates within the forecasting models. The resulting subset of weather indicators represents a balance between capturing meaningful environmental signals and maintaining a parsimonious model specification.

Following feature engineering and multicollinearity filtering, the selected weather variables were integrated into the forecasting dataset as potential exogenous regressors for the SARIMAX models. These variables were aligned temporally with the monthly retail demand data and prepared for inclusion in the forecasting framework. The engineered features therefore provide the foundation for evaluating whether weather-sensitive predictors can enhance the explanatory power and predictive accuracy of the demand forecasting models.

Figure 3. *Correlation heatmap of candidate weather variables used in the feature selection process*



3.3.Sector Selection Strategy

The retail dataset includes twenty-eight sectors with varying demand characteristics and sensitivity to external factors. Modeling all sectors would introduce unnecessary complexity while providing limited insight into the specific objective of evaluating weather-sensitive forecasting. To focus the analysis, a subset of sectors was selected based on their observed relationship with weather variables.

A Cross-Correlation Function (CCF) analysis was conducted to measure the relationship between retail demand and key weather indicators across different time lags. This approach allows identification of both immediate and delayed associations between weather conditions and demand patterns. For each sector, the strength of correlation between demand and weather variables was evaluated, and sectors were ranked based on the magnitude of these relationships.

Sectors exhibiting stronger correlations with weather variables were considered more likely to benefit from the inclusion of weather in forecasting models. Based on this ranking, five sectors were selected for detailed analysis (Table 1). This data-driven approach ensures that the evaluation of weather-sensitive models is conducted in contexts where weather effects are more likely to be present and measurable.

Table 1. *Final 5 selected sectors based on CCF results.*

Rank	Sector	Max CCF	Feature	Lag
1	Other motor vehicle dealers [4412]	0.219	CDD_18	0
2	Convenience retailers and vending machine operators [44513]	0.157	CDD_18	0
3	Miscellaneous retailers [459B]	0.149	CDD_18	2
4	Electronics and appliances retailers [4492]	0.140	CDD_18	3
5	New car dealers [44111]	0.139	CDD_18	3

This targeted selection improves the relevance of the analysis and allows for a clearer assessment of whether incorporating weather variables leads to meaningful improvements in forecasting performance.

3.4. Time Series Diagnostics

Before estimating the forecasting models, time series diagnostics were performed to better understand the underlying structure of the retail demand data. Retail sales data often exhibit strong seasonal patterns driven by recurring consumer behavior cycles such as holidays, seasonal product demand, and broader economic activity. If these patterns are not properly accounted for, forecasting models may incorrectly attribute regular seasonal fluctuations to external variables such as weather conditions (Saputra & Kumar, 2024). Therefore, identifying and separating the different structural components of the demand series was an important step in preparing the data for statistical modeling.

To address this, **Seasonal-Trend Decomposition using Loess (STL)** was applied to the selected retail sectors. STL separates a time series into three components: a long-term trend, a recurring seasonal pattern, and a remainder (residual) component. In this study, the analysis focuses specifically on the residual component, which represents demand variation not explained by trend or deterministic seasonality.

This step was particularly important because the analysis is based on monthly retail data, which contains strong built-in seasonal effects. Ideally, weather-driven demand modeling would be conducted using daily demand data to capture short-term responses to weather conditions. However, due to the lack of publicly available daily retail sales data, the analysis is performed at a monthly level.

By isolating and using the residual component of the demand series, the study reduces the influence of predictable seasonal cycles and long-term trends. This allows the forecasting models to focus on irregular fluctuations in demand, where the impact of external factors such as weather conditions is more likely to be present.

3.5. Forecasting Methodology

Forecasting was carried out using four models: **Naïve**, **Seasonal Naïve**, **SARIMA**, and **SARIMAX**. The Naïve and Seasonal Naïve models were included as benchmark methods to test whether the more advanced models provide meaningful improvement beyond simple persistence rules. The Naïve model uses the most recent observed value as the next forecast, while the Seasonal Naïve model uses the value from the same month in the previous year.

For the statistical models, SARIMA was used as the baseline model and SARIMAX as the weather-sensitive extension. Both were built on the same differencing structure, with $d = 1$ and $D = 1$ fixed across sectors. Model orders were then selected separately for each chosen sector using

an AIC-based grid search over candidate values of $p, q \in \{0,1,2\}$ and $P, Q \in \{0,1\}$, with seasonal period $s = 12$. The best order for each sector was identified using the lowest AIC, and these choices were then checked against the ACF and PACF patterns of the differenced series. This process showed that the selected sectors consistently favored autoregressive structure with no moving average terms.

Once the best order was selected for a sector, that same SARIMA structure was used for both the SARIMA and SARIMAX versions of the model. This was an important design choice, because it ensures that any difference in forecast accuracy is due to the weather regressors rather than changes in ARIMA specification.

Weather regressors for SARIMAX were selected separately for each sector using cross-correlation analysis between STL residuals and weather variables at lags 0 to 3 months. A weather feature was retained if its maximum absolute cross-correlation reached at least 0.05 at any lag. This threshold was used to filter weak noise while preserving modest but potentially meaningful residual weather signals. The selected lag for each feature was the lag at which that maximum relationship occurred. This allowed the model to capture both immediate weather effects and delayed responses, depending on the sector. Notably, Heating Degree Days (HDD) did not meet the threshold for any selected sector, indicating that its effect was already absorbed by the seasonal structure of the series.

All models were implemented in Python. The SARIMA and SARIMAX models were estimated using the statsmodels library through maximum likelihood estimation. Forecast accuracy was evaluated using a rolling-origin expanding-window design. Each model was first trained on a minimum of 60 months of data, then used to forecast the next month. The training

window was expanded one month at a time, and the process was repeated until the end of the series, producing 47 out-of-sample forecasts per sector.

For SARIMAX, weather regressors were standardized within each training window using only the training-period mean and standard deviation. This prevented data leakage from the test period. Lagged regressors were aligned so that a lag-k weather variable at time $t$used weather information from time $t - k$.

This framework provides a controlled comparison across simple benchmarks, traditional time-series models, and weather-sensitive models under realistic out-of-sample forecasting conditions.

3.6. Forecast Evaluation and Results

Forecast performance was evaluated using a rolling-origin expanding-window approach designed to reflect realistic forecasting conditions. Each model was initially trained on 60 months of data and then used to predict the following month. After each forecast, the training window was expanded by one month and the process was repeated until the end of the series. This produced 47 out-of-sample forecasts per sector, allowing model performance to be assessed over multiple time periods rather than a single train-test split.

Model accuracy was measured using Mean Absolute Percentage Error (MAPE) as the primary evaluation metric, supported by Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAPE was treated as the main criterion because it allows forecast errors to be interpreted relative to the scale of demand, making comparisons across sectors more meaningful (Table 2).

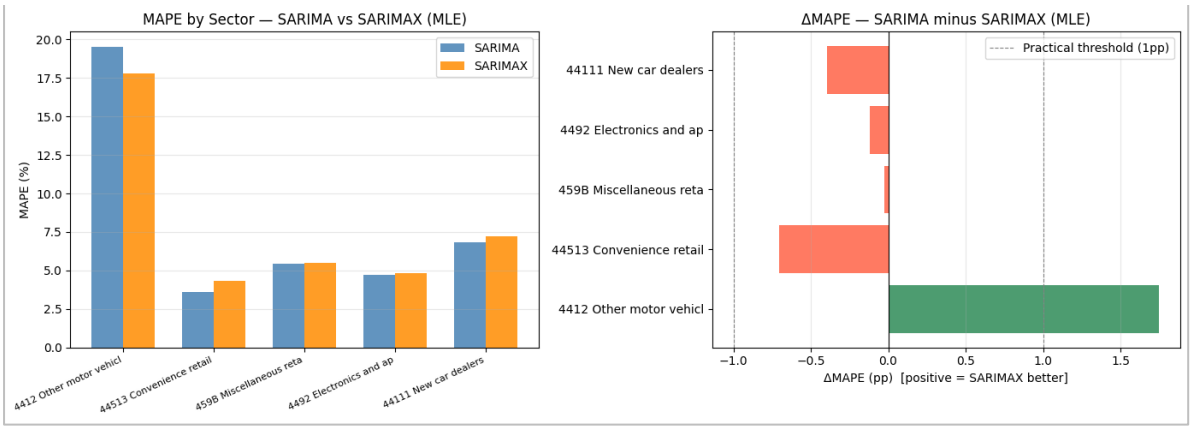
Table 2. *Sector-level comparison of SARIMA and SARIMAX Results Resume Table*

Sector	SARIMA MAPE	SARIMAX MAPE	ΔMAPE (pp)	Diebold-Mariano p-value	Practical Significance (≥1 pp)
Other motor vehicle dealers [4412]	19.53%	17.78%	1.75	0.042	Yes
Convenience retailers [44513]	3.62%	4.32%	-0.7	>0.05	No
Miscellaneous retailers [459B]	5.46%	5.48%	-0.02	>0.05	No
Electronics and appliances [4492]	4.69%	4.81%	-0.12	>0.05	No
New car dealers [44111]	6.82%	7.22%	-0.39	>0.05	No

To assess practical significance, an improvement of at least 1 percentage point in MAPE was treated as a meaningful gain in forecasting performance.

The results show that weather variables did not improve forecasting performance uniformly across sectors. Among the five selected sectors, the SARIMAX model meaningfully outperformed SARIMA in only one sector: Other motor vehicle dealers [4412] (Figure 4). In this case, MAPE improved from 19.53% under SARIMA to 17.78% under SARIMAX, indicating that weather variables contributed useful predictive information.

Figure 4. *Sector-level comparison of SARIMA and SARIMAX forecast accuracy using MAPE.*



For the remaining four sectors, the inclusion of weather variables did not lead to better forecasts. In Convenience retailers and vending machine operators [44513], SARIMA produced a lower MAPE than SARIMAX, at 3.62% versus 4.33%. In Miscellaneous retailers [459B], the difference was minimal, but SARIMA still performed slightly better, at 5.46% versus 5.48%.

The same pattern appeared in Electronics and appliances retailers [4492], where SARIMA achieved 4.69% compared with 4.81% for SARIMAX, and in New car dealers [44111], where SARIMA recorded 6.82% compared with 7.22% for SARIMAX.

The difference between SARIMA and SARIMAX was also tested using the Diebold-Mariano test. The results showed that the improvement for *Other motor vehicle dealers [4412]* was statistically significant, while the differences in the other four sectors were not. This indicates that weather added measurable predictive value only for that sector.

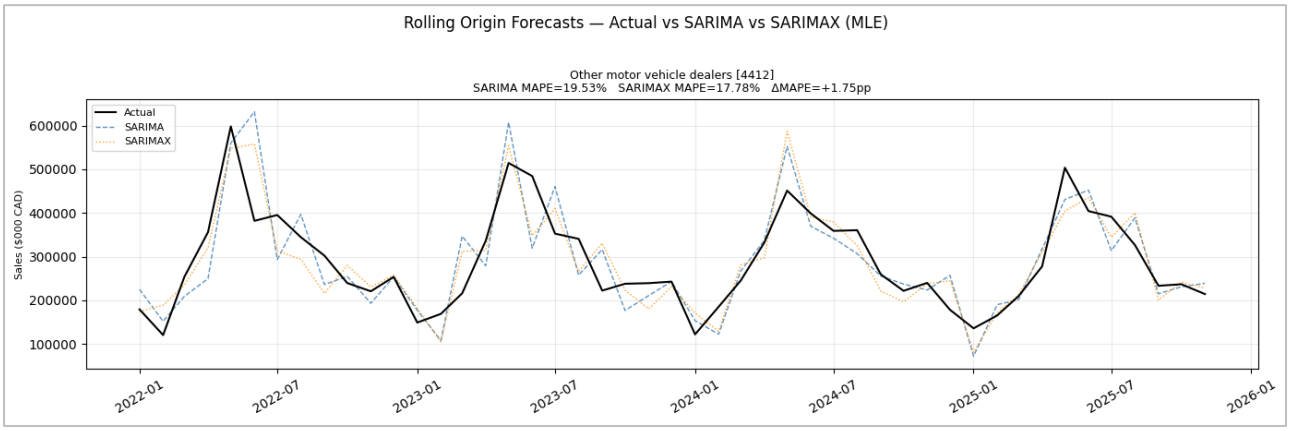
These findings suggest that weather sensitivity is sector-specific rather than generalizable across retail demand. Although the selected sectors were chosen based on their relative relationship with weather, only one sector showed a practically meaningful forecasting improvement after weather variables were introduced. This indicates that even when weather signals are present, their predictive value may be limited once the underlying trend and seasonal structure of the series have already been modeled.

The benchmark models also provide useful context for interpreting performance. Including Naïve and Seasonal Naïve models confirmed that the statistical models were being evaluated against simple alternatives rather than in isolation. The stronger performance of SARIMA and SARIMAX relative to these benchmarks supports the use of time-series modelling, while the comparison between SARIMA and SARIMAX isolates the added value of weather information.

Model reliability was further supported by the convergence diagnostics recorded during estimation. Across all sectors, models, and rolling forecast iterations, the forecasting pipeline completed without convergence failures, resulting in zero failures across 470 estimation attempts. This indicates that the reported results are computationally stable and not driven by estimation breakdowns.

Overall, the forecasting results provide partial support for the first research question. Weather variables improved demand forecasting accuracy in a meaningful way for Other motor vehicle dealers [4412], but not for the other four selected sectors. This suggests that the value of weather-sensitive forecasting depends on sector characteristics and should be applied selectively rather than assumed to benefit all retail categories (Figure 5).

Figure 5. *Example forecast plot showing actual values versus model predictions for Other motor vehicle dealers [4412].*



4. Workforce Allocation Optimization (Phase 2)

4.1. From Forecasting to Workforce Allocation

The second phase of the project evaluates whether improved demand forecasts lead to more efficient workforce allocation. While Phase 1 focused on forecast accuracy, Phase 2 translates those forecasts into staffing decisions and compares the operational outcomes under traditional and weather-sensitive forecasting. The goal is to determine whether a more accurate forecast also reduces staffing mismatch in practice.

The analysis uses the rolling one-step-ahead forecasts generated in Phase 1 for the same five selected sectors. For each evaluation month, staffing is calculated twice: once using the SARIMA forecast and once using the SARIMAX forecast. These forecast-based staffing

decisions are then compared against the staffing level that would have been required based on actual realized demand.

4.2. Demand-to-Labor Conversion

To convert forecasted sales into workforce requirements, the study derives an empirical labor-share parameter, denoted by λ, using public labor market data from Statistics Canada (Appendix A). Two monthly provincial series were used for this purpose: employment in wholesale and retail trade and average weekly earnings in trade. These series were matched with aggregate Ontario retail sales across all 28 retail sectors to estimate the monthly labor-cost-to-sales ratio.

The monthly ratio was calculated as:

$$\lambda_t = \frac{N_t \times w_t \times 4.33}{S_t^{all}}$$

where $N_t$is monthly employment, $w_t$is average weekly earnings, 4.33 converts weekly wages to monthly wages, and $S_t^{all}$is total retail sales across all sectors. Using all 28 sectors in the sales denominator was necessary to match the scope of the employment data and avoid overstating labor intensity. This step was included to derive realistic productivity-based staffing estimates that translate forecasted sales into interpretable workforce requirements. Although the exact parameter values could not be accurate due to aggregations, this does not affect the comparative validity of the analysis because both forecasting models are evaluated using the same conversion framework. As a result, any differences in workforce allocation outcomes can be primarily attributed to differences in forecast performance rather than to the conversion parameters themselves.

To reduce distortion from abnormal pandemic conditions, the final labor share used in the optimization model was based on the COVID-excluded average. This produced a labor share of

9.45%, which is close to the industry benchmarks. The average monthly wage was estimated at $3,857.52 per employee.

Using these parameters, forecasted sales were converted into forecast-based staffing allocations, and actual sales were converted into required staffing levels. In simplified form, the staffing rule is:

$$e_t = \text{round}\left(\frac{\text{Forecast}_t \times 1000 \times \lambda}{w}\right)$$

where e_t is the number of employees allocated for month t, λ is the labor share, and w is the monthly wage. The same formula was applied to actual sales to determine the workforce level that would have matched realized demand.

4.3. Optimization Model

The workforce allocation model was implemented as a linear programming framework that minimizes staffing mismatch cost for each sector and month. The model compares the workforce implied by the forecast with the workforce actually required and then assigns a penalty to the resulting overstaffing or understaffing.

Mismatch was defined using two components:

- **overstaffing**: allocated employees minus required employees.
- **understaffing**: required employees minus allocated.

These two outcomes were not treated equally. Following Mani et al. (2015), understaffing was assigned a penalty twice as large as overstaffing, reflecting the higher operational risk associated with insufficient staffing. As a result, the monthly penalty costs were defined as:

overstaffing cost = monthly wage per employee-month

understaffing cost = 2 × monthly wage per employee-month

Using the estimated wage value, the overstaffing cost was $3,857.52 per employee-month and the understaffing cost was $7,715.04 per employee-month.

For each evaluation window, the LP therefore minimized total mismatch cost subject to the staffing decision implied by the forecast. In practice, this means that a lower forecast error translates into a staffing level closer to the level actually needed, which in turn reduces both mismatch and penalty cost. The optimization was repeated for every rolling forecast month and for both forecasting scenarios, creating a direct comparison between workforce allocation based on SARIMA and workforce allocation based on SARIMAX.

4.4. Scenario Comparison and Evaluation

Two staffing scenarios were compared for each sector: staffing based on SARIMA forecasts, and staffing based on SARIMAX forecasts. The evaluation was conducted over the same 47 rolling forecast windows used in Phase 1. For each sector, total overstaffing, total understaffing, and total mismatch were aggregated across all employee-months. In addition, the associated penalty costs were summed across the full evaluation period.

The primary outcome in this phase was total workforce mismatch, measured in employee-months. This directly captures how far forecast-based staffing decisions deviated from actual workforce requirements. Penalty cost was treated as a secondary outcome, providing a monetary interpretation of the mismatch.

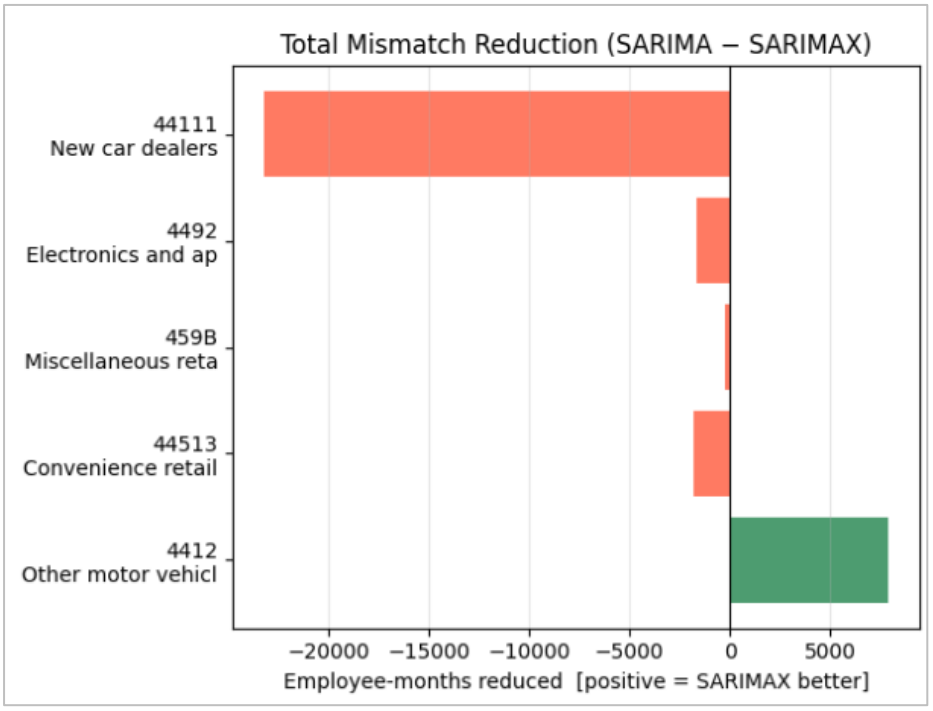
To assess whether SARIMAX consistently reduced mismatch relative to SARIMA over time, the monthly mismatch differences were also tested using the Wilcoxon signed-rank test. This nonparametric test was preferred over a paired t-test because the mismatch difference series may contain outliers and may not satisfy normality assumptions.

4.5. Results

The results show that workforce allocation outcomes followed the forecast accuracy results from Phase 1. Weather-sensitive forecasting improved workforce allocation only in the one sector where it had also improved forecast accuracy: Other motor vehicle dealers [4412].

For that sector, SARIMAX reduced total workforce mismatch from 62,985 to 55,098 employee-months, a reduction of 7,887 employee-months, or 12.52% (Figure 6). The same pattern appeared in cost terms, where total penalty cost fell from $362.93 million under SARIMA to $320.75 million under SARIMAX, producing a saving of $42.18 million, or 11.62%.

Figure 6. *Total workforce mismatch by sector under SARIMA and SA*



In the other four sectors, SARIMAX did not improve allocation outcomes. These results indicate that when weather does not improve the forecast itself, it also does not improve downstream staffing decisions.

The statistical test further supports a cautious interpretation. Although SARIMAX produced the best workforce outcome for Other motor vehicle dealers [4412], the Wilcoxon signed-rank test did not show a statistically significant monthly mismatch reduction at the 5% level. The same was true for the other sectors, where no significant difference was detected. This means that the operational advantage of weather-sensitive forecasting was observable in aggregate for one sector, but not strong enough to establish consistent monthly superiority across the evaluation period.

Overall, the second research question receives partial support. Weather-sensitive forecasts can improve workforce allocation efficiency, but only when they first provide a meaningful improvement in forecast accuracy. In this analysis, that occurred in one sector only. As a result, the value of weather-informed staffing appears to be conditional rather than universal: it is most useful in sectors where demand responds to weather in a measurable way.

5. Interactive Dashboard

To enhance interpreatability and allow dynamic exploration of the results an interactive dashborad was developed in Power Bi. The dashboard integrates outputs from both phase 1 and phase 2, enabling users to filter by sector, time period and forecasting model.

The key views of the dashboard are presented below:

Figure 7.1. *Power BI dashboard – Forecast Accuracy Overview*

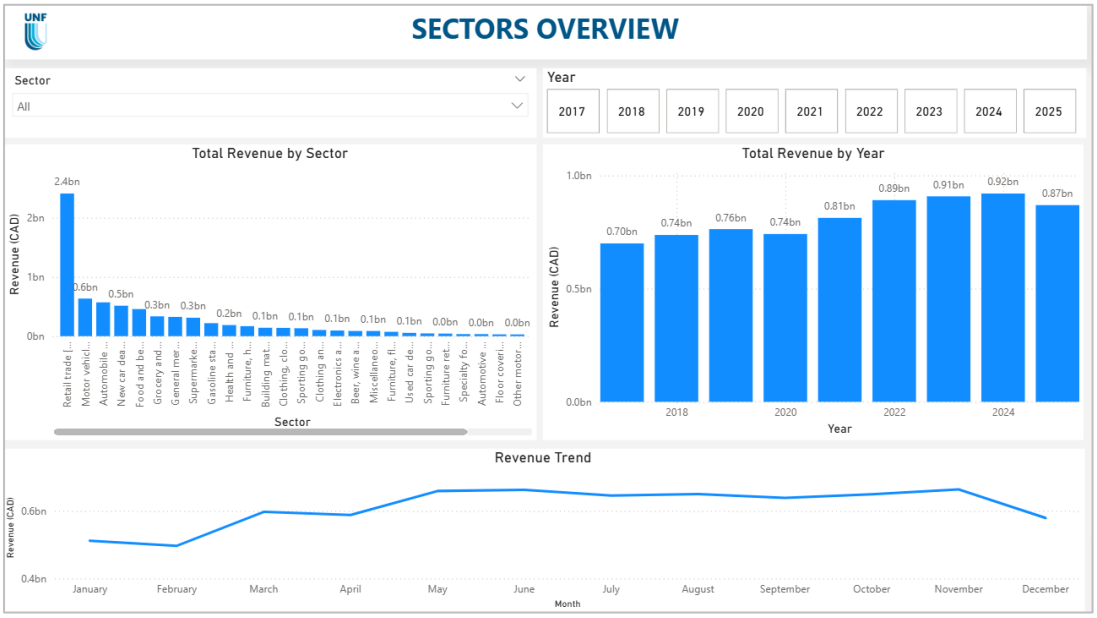
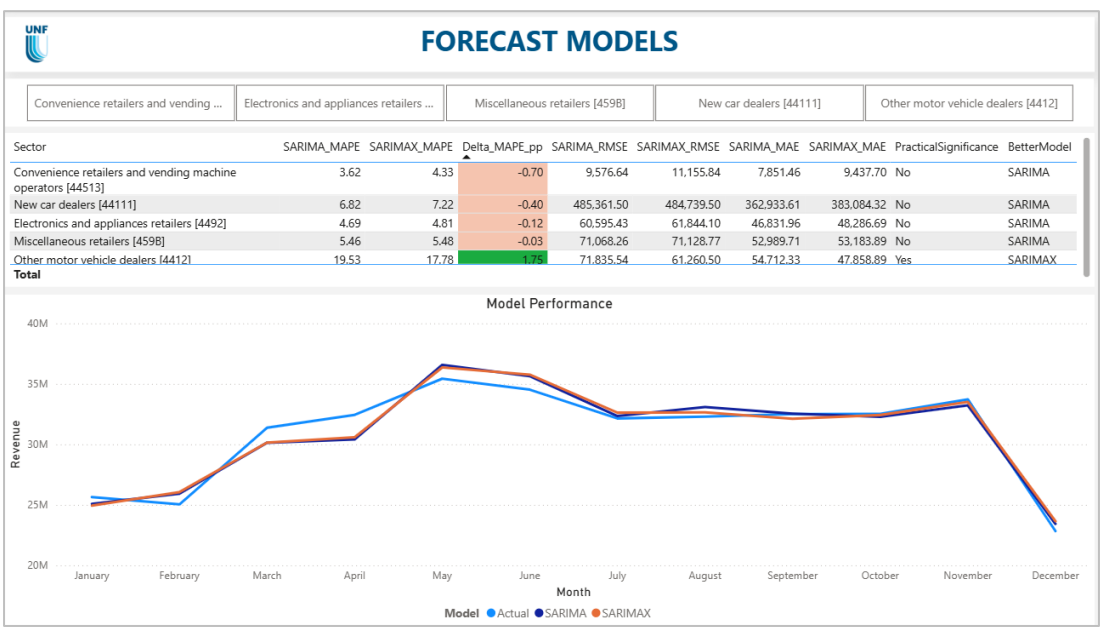
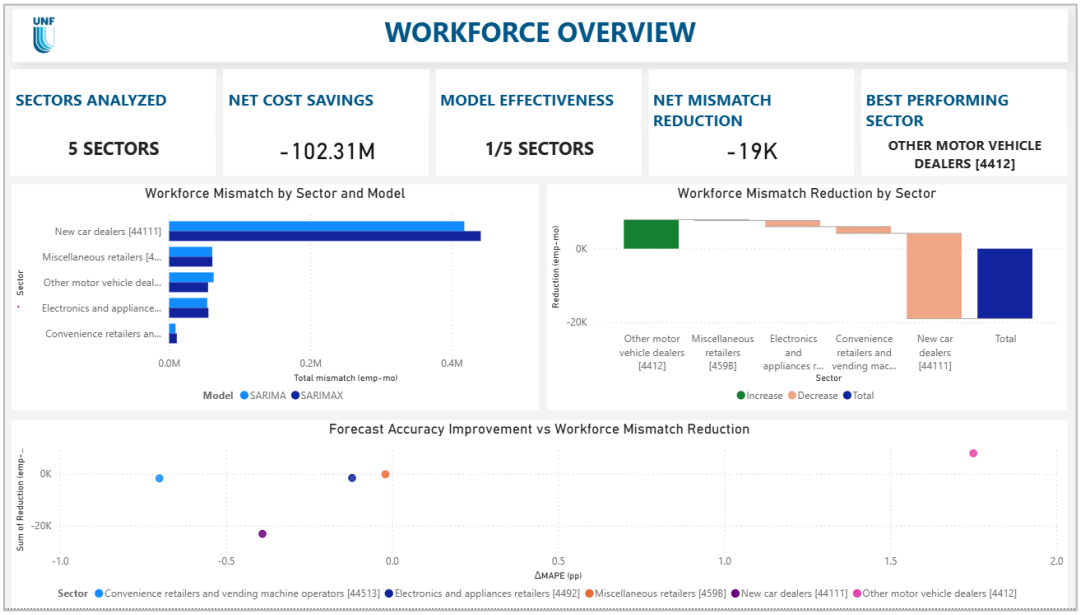


Figure 7.2. *Power BI dashboard – Time-Series Forecast Comparison*



Sector	SARIMA_MAPE	SARIMAX_MAPE	Delta_MAPE_pp	SARIMA_RMSE	SARIMAX_RMSE	SARIMA_MAE	SARIMAX_MAE	PracticalSignificance	BetterModel
Convenience retailers and vending machine operators [44513]	3.62	4.33	-0.70	9,576.64	11,155.84	7,851.46	9,437.70	No	SARIMA
New car dealers [44111]	6.82	7.22	-0.40	485,361.50	484,739.50	362,933.61	383,084.32	No	SARIMA
Electronics and appliances retailers [4492]	4.69	4.81	-0.12	60,595.43	61,844.10	46,831.96	48,286.69	No	SARIMA
Miscellaneous retailers [459B]	5.46	5.48	-0.03	71,068.26	71,128.77	52,989.71	53,183.89	No	SARIMA
Other motor vehicle dealers [4412]	19.53	17.78	1.75	71,835.54	61,260.50	54,712.33	47,858.89	Yes	SARIMAX
Total									

Figure 7.3. *Power BI dashboard – Workforce Mismatch and Cost Reduction*



The interactive dashboard provides a comprehensive overview of both phases of the project, it allows users to visually analize the performance of the SARIMA AND SARIMAX models side by side and observe how weather variables affect forecast accuracy across sectors.

6. Findings and Discussion

6.1. Key Findings

The results of this study show that weather information does not improve retail demand forecasting uniformly across sectors. In Phase 1, the weather-sensitive SARIMAX model outperformed the baseline SARIMA model in only one of the five selected sectors: Other motor vehicle dealers [4412]. For the other four sectors, including weather variables did not improve forecast accuracy and in some cases slightly worsened it. This suggests that the predictive value of weather is limited and depends on sector-specific demand behavior rather than applying broadly across retail activity.

The workforce allocation results followed the same pattern. In Phase 2, staffing decisions based on weather-sensitive forecasts reduced total workforce mismatch and penalty cost only for

Other motor vehicle dealers [4412]. In the remaining sectors, the weather-sensitive forecasts did not produce better operational outcomes. This indicates that improvements in workforce allocation depend on improvements in forecast accuracy first; when weather does not enhance forecasting, it also does not enhance staffing decisions.

Taken together, the findings provide partial support for both research questions. Weather-sensitive forecasting can improve demand prediction and workforce allocation efficiency, but only in sectors where demand responds to weather in a measurable way.

6.2. Discussion

The results suggest that weather-sensitive forecasting has limited value when applied to aggregated monthly retail demand. Although weather variables improved forecasting and workforce allocation for Other motor vehicle dealers [4412], the same effect did not appear in the other selected sectors.

One important explanation is the level of data aggregation used in the study. Weather effects are often short-term and event-driven, while monthly retail sales data already embeds strong seasonal patterns. As a result, it becomes difficult to distinguish the effect of specific weather conditions from broader seasonal demand cycles. This limitation likely reduces the visibility of weather signals in the forecasting models, even when such effects exist in practice.

The results also show that improvements in workforce allocation are conditional on improvements in forecast accuracy. The weather-sensitive model only improved staffing outcomes in the same sector where it improved forecasting performance. This reinforces the idea that workforce optimization does not create value independently; it depends on the quality of the forecasts used as inputs.

Despite the limited sector-wide forecasting gains, the study demonstrates a complete analytical framework for integrating weather data into demand forecasting and workforce allocation. This is an important contribution because it shows how external environmental variables can be translated into operational staffing decisions, even if the measurable impact depends heavily on data granularity and sector characteristics.

Overall, the findings support a selective rather than universal use of weather-informed forecasting. Weather variables may provide operational value in sectors with stronger weather sensitivity, but they should not be assumed to improve performance across all retail contexts.

6.3. Limitations

The main limitation of this study is the lack of publicly available daily retail demand data. While weather conditions were available at a daily level, the retail sales data used in the analysis was only available monthly. This required the study to aggregate weather information to the monthly level, which reduces the ability to isolate short-term weather effects from regular seasonal demand patterns.

This limitation is especially important because weather effects are often immediate and event-based. Rainfall, snowfall, or unusual temperature changes may influence consumer behavior over a few days rather than across an entire month. When demand is observed only at the monthly level, these short-term effects are blended with broader seasonal patterns, making them harder to detect statistically.

A second limitation is that the workforce allocation model depends on estimated labor-share and wage parameters derived from aggregate public data. Although these values provide a realistic basis for translating forecasts into staffing levels, they remain approximations and do not represent firm-level productivity or scheduling practices. However, because both forecasting

models were evaluated using the same conversion framework, this does not affect the comparative validity of the results.

Despite these limitations, the study still demonstrates a complete analytical framework linking weather data, demand forecasting, and workforce allocation. The findings should therefore be interpreted as evidence of how the methodology performs under public-data constraints, rather than as a final statement on the full operational value of weather-sensitive forecasting.

7. Conclusion and Recommendations

7.1. Conclusion

This study examined whether weather variables improve retail demand forecasting accuracy and whether those improvements translate into more efficient workforce allocation. Using monthly Ontario retail sales data and aggregated weather data, the project compared traditional forecasting models with weather-sensitive alternatives across five selected retail sectors.

The results showed that weather variables improved forecasting performance in only one sector. The same sector also showed lower workforce mismatch and lower staffing penalty cost when weather-sensitive forecasts were used. In the remaining sectors, weather variables did not improve forecasting or staffing outcomes.

Overall, the study provides partial support for both research questions. More importantly, it demonstrates a complete analytical framework for integrating weather data into demand forecasting and linking those forecasts to workforce planning. The results suggest that the value of weather-sensitive forecasting is conditional on sector characteristics and data granularity rather than universally applicable across retail demand.

7.2. Recommendations

Organizations should apply weather-sensitive forecasting selectively rather than assuming it will improve performance in all retail sectors. Before adopting such models operationally, firms should first test whether their demand shows a measurable relationship with weather variables.

Where weather sensitivity is present, forecast outputs should be linked directly to staffing and workforce planning systems so that predictive improvements can translate into operational benefits. This is especially relevant for sectors where demand responds to short-term changes in environmental conditions.

For researchers and analysts, future implementations of this framework should prioritize higher-frequency demand data whenever possible. Daily or transaction-level sales data would provide a stronger basis for isolating weather effects and would likely produce a more accurate test of the methodology.

7.3. Future Research

Future research should repeat this analysis using daily retail demand data or firm-level operational data to better capture short-term weather effects. Additional studies could also test the framework in other industries, such as food services, transportation, or delivery operations, where weather may have a stronger and more immediate influence on demand.

The forecasting stage could also be extended by comparing traditional time-series models with machine learning approaches using the same weather variables. On the workforce side, future work could incorporate additional operational constraints such as shift length, part-time staffing, scheduling flexibility, and service-level requirements to produce a more detailed workforce allocation model.

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Appendix A — Raw Data Sources

Dataset 1: ontario_daily_weather_2017_2025

Source: NASA POWER

URL: https://power.larc.nasa.gov (data collected using python API) *

Time Period: January 2017 – December 2025

Frequency: Daily

Variables downloaded:

Variable	Definition
Date	Date of the recorded weather observation.
Temp_Mean	Average daily air temperature at 2 meters above ground.
Temp_Max	Highest daily air temperature at 2 meters above ground.
Temp_Min	Lowest daily air temperature at 2 meters above ground.
Humidity	Average daily relative humidity.
Dew_Point	Dew point temperature; indicates how muggy or saturated the air is.
Precipitation	Total daily precipitation, including rain and melted snow.
Snowfall	Daily precipitation that fell as snow, measured as melted-water equivalent.
Derived (Rainfall_Est)	Estimated rainfall, calculated as total precipitation minus snowfall.
Snow_Depth	Depth of snow accumulated on the ground.
Solar_Level	Total daily solar energy reaching the ground.
Cloud_Amount	Percentage of sky covered by clouds during the day.
Wind_Speed	Average daily wind speed at 10 meters above ground.

Dataset 2: Monthly retail trade sales by industry

Source: Statistics Canada

Table: 20-10-0056-02

URL: https://doi.org/10.25318/2010005601-eng

Time Period: January 2017 – November 2025

Frequency: Monthly

Variables downloaded:

Variable	Definition
North American Industry Classification System (NAICS)	The names and codes of all trade industries following NAICS naming conventions
Sales (x The Number of Months)	There is a column for each of the months included in this analysis, each cell in the column shows the sales of the correspondin sector (x $1,000).

Dataset 3: Employment by industry

Source: Statistics Canada

Table: 14-10-0355-01

URL: https://doi.org/10.25318/1410035501-eng

Time Period: January 2017 – December 2025

Frequency: Monthly

Variables downloaded:

Variable	Definition
North American Industry Classification System (NAICS)	The names and codes of all trade industries following NAICS naming conventions
Number of Employees (x The Number of Months)	There is a column for each of the months included in this analysis, each cell in the column shows the number of employees in the correspondin sector in that month.

Dataset 3: Average weekly earnings by industry

Source: Statistics Canada

Table: 14-10-0203-01

URL: https://doi.org/10.25318/1410020301-eng

Time Period: January 2017 – December 2025

Frequency: Monthly

Variable	Definition
North American Industry Classification System (NAICS)	The names and codes of all trade industries following NAICS naming conventions
Wages (x The Number of Months)	There is a column for each of the months included in this analysis, each cell in the column shows the average weekly wages of employees in the correspondin sector in that month.